Lumiq: AI-Powered Investment Analysis Platform

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Course: COMSW4995 - Applied Machine Learning

Project Resources:
🌐 Live Application: https://huggingface.co/spaces/Shruti02222/ai-investment-analyst
💻 GitHub Repository: https://github.com/Shruti022/Finance_chatbot
📂 Deployment Code: https://huggingface.co/spaces/Shruti02222/ai-investment-analyst/tree/main
🎥 Project Demo Video:
https://drive.google.com/file/d/1aaqKMaz4g4eZJ-baMMCrdd9LcTU0twd_/view?usp=sharing
📑Slides: https://www.canva.com/design/DAG7_ck60UY/8K0ZixTOtkdBkD930YgTKg/edit?utm_content=DAG7_ck60UY&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

ABSTRACT

Lumiq is a multi-agent AI investment platform that democratizes institutional-grade financial analysis for retail investors and students. The system employs six specialized AI agents that analyze stocks from complementary perspectives: fundamental valuation, market trends, risk assessment, and technical patterns. Each agent uses retrieval-augmented generation to ground analysis in a curated knowledge base of 128 financial documents plus live market data. Users receive structured BUY/SELL/HOLD recommendations with conviction scores (1-10) and specific action plans including entry prices, stop-loss levels, and timelines. Additionally, the platform supports automated financial statement analysis by parsing uploaded PDF, Excel, or CSV documents to extract key metrics and generate interpretive reports. The multi-agent architecture ensures balanced decision-making by synthesizing independent assessments before delivering final recommendations, making professional-quality investment analysis accessible for educational purposes.

INTRODUCTION

The Problem

Investing requires analyzing company fundamentals, market trends, risk factors, and technical patterns simultaneously. Retail investors and students face major barriers: Bloomberg Terminal costs $20,000+ annually, free tools like Yahoo Finance provide raw data without guidance, and generic chatbots lack structured analyst perspectives or concrete investment recommendations.

Our Solution

Lumiq employs six specialized AI agents that independently analyze stocks and assign sentiment scores (bullish +1, neutral 0, bearish -1). A Chief Investment Officer synthesizes these into final BUY/SELL/HOLD recommendations with conviction ratings and specific action plans. A Financial Statement Analyst parses uploaded documents (PDF/Excel/CSV) to extract metrics and generate structured reports. Each agent uses retrieval-augmented generation with 128 curated financial documents, grounding recommendations in historical patterns and current data.

Who Benefits

Lumiq makes professional analysis accessible to students, researchers, and underrepresented groups (including women historically excluded from finance spaces) by providing clear, structured explanations that demystify investment decisions. Our corporate experience analyzing lengthy 10-K reports inspired us to automate this time-consuming document parsing process.

Project Goals

Lumiq delivers: (1) Educational accessibility through free multi-perspective analysis, (2) Financial inclusivity via approachable interfaces, (3) Technical demonstration of RAG and multi-agent architectures in finance, and (4) Actionable recommendations with price targets and timelines. All outputs include disclaimers to consult qualified advisors before investing.

RELATED WORK

RAG Systems in Finance

Retrieval-augmented generation has been adopted in financial applications to enhance LLM accuracy. The CFA Institute demonstrated RAG for investment document analysis, while commercial platforms like Auquan deploy RAG-based systems for real-time trading signals. However, existing implementations focus on single-task workflows (document summarization, Q&A) rather than multi-perspective analysis, and remain proprietary and expensive.

Multi-Agent AI Systems

Multi-agent architectures have been explored for financial forecasting. TradingAgents (2024) demonstrates collaborative market analysis with specialized roles, while recent arXiv research (2025) proposed multi-agent systems analyzing news sentiment, technical indicators, and macroeconomic factors. These systems prioritize automated trading execution with opaque agent interactions, making it difficult for users to understand decision-making processes.

Financial Statement Analysis Tools

Automated financial statement analysis tools include S&P Global's NLP solutions for extracting structured data, V7 Labs' computer vision research (2024) for parsing balance sheets, and BloombergGPT for document summarization. These excel at data extraction but operate as black-box services that return metrics without interpretive guidance on investment implications.

Lumiq's Positioning

Table 1: Competitive Analysis - Lumiq's Positioning

Feature	Bloomberg / FactSet	Yahoo Finance	ChatGPT	Lumiq
Cost	$20,000–$40,000/yr	Free	Free	Free
Multi-Agent Analysis	✓ (human analysts)	✗	✗	✓
Structured Recommendations	✓	✗	✗	✓ (BUY/SELL/HOLD)
Document Parsing	✓	✗	Partial	✓ (PDF/Excel/CSV)
Interpretive Reports	✓	✗	✗	✓ (6-section analysis)
Educational Transparency	✗	✗	✗	✓ (visible agent reasoning)

Lumiq uniquely combines professional-grade multi-perspective rigor with educational accessibility, making it valuable for academic contexts where cost constraints limit access to institutional tools.

METHODOLOGY

5.1 System Architecture

Lumiq integrates three core components shown in **Figure 1**: a RAG pipeline for knowledge retrieval, a multi-agent system for collaborative analysis, and a document parser for financial statements.

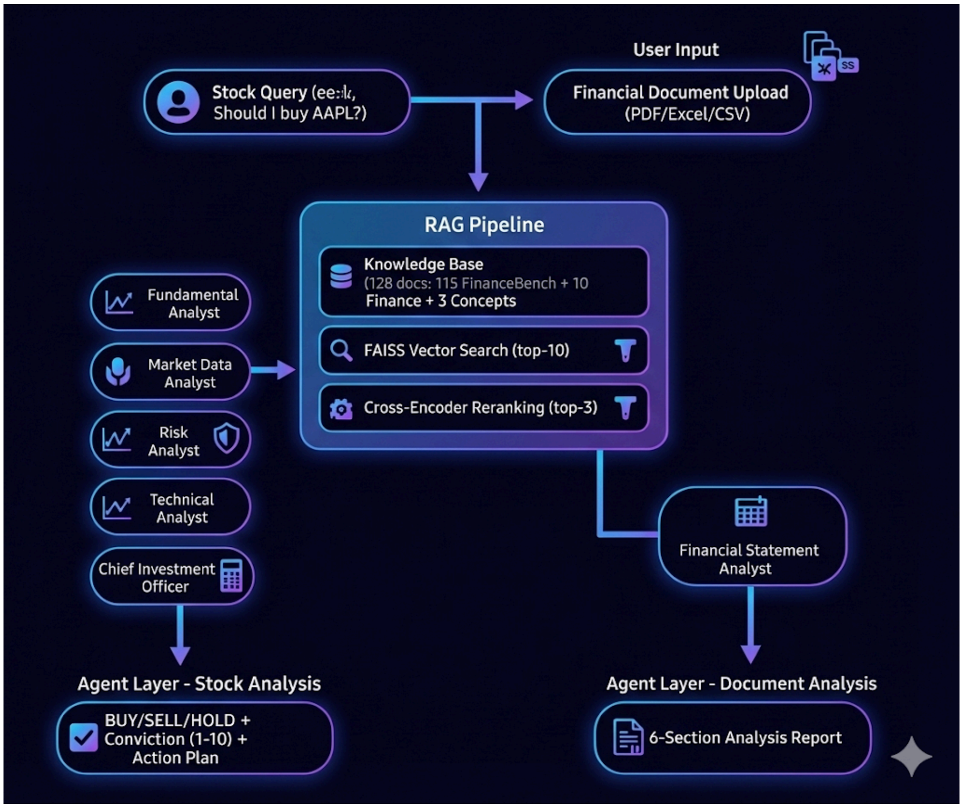

Figure 1: System Architecture - RAG Pipeline and Multi-Agent Framework

5.2 Stock Analysis Workflow

The stock analysis process follows five stages illustrated in **Figure 2**: ticker extraction, parallel specialist agent analysis with sentiment scoring, Chief Investment Officer synthesis, and final recommendation with conviction rating.

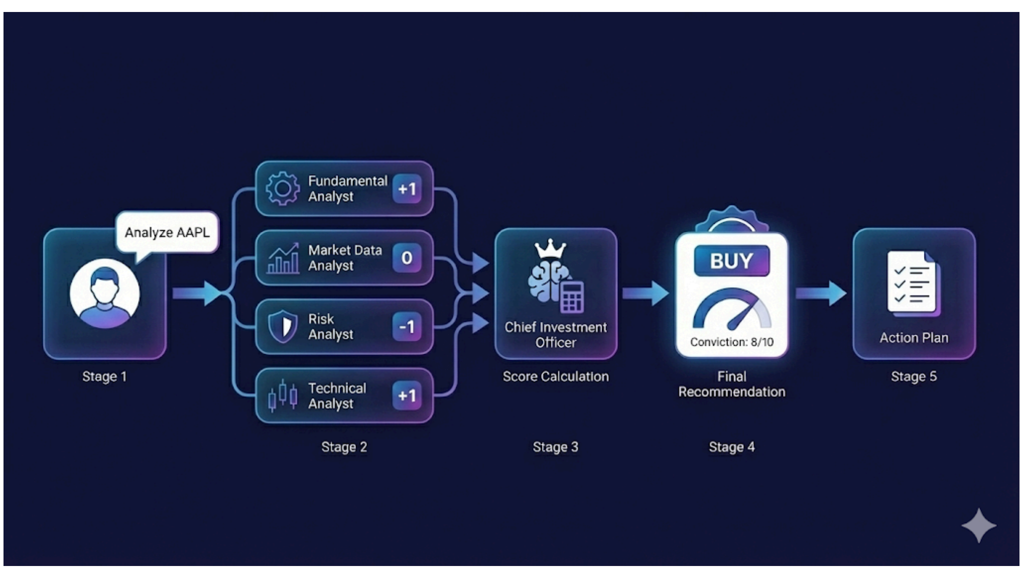

Figure 2: Multi-Agent Stock Analysis Workflow

5.3 RAG Pipeline

The two-stage retrieval process shown in **Figure 3** uses FAISS vector search to retrieve 10 candidates from 128 documents, then cross-encoder reranking selects the top 3 contexts for injection into agent prompts.

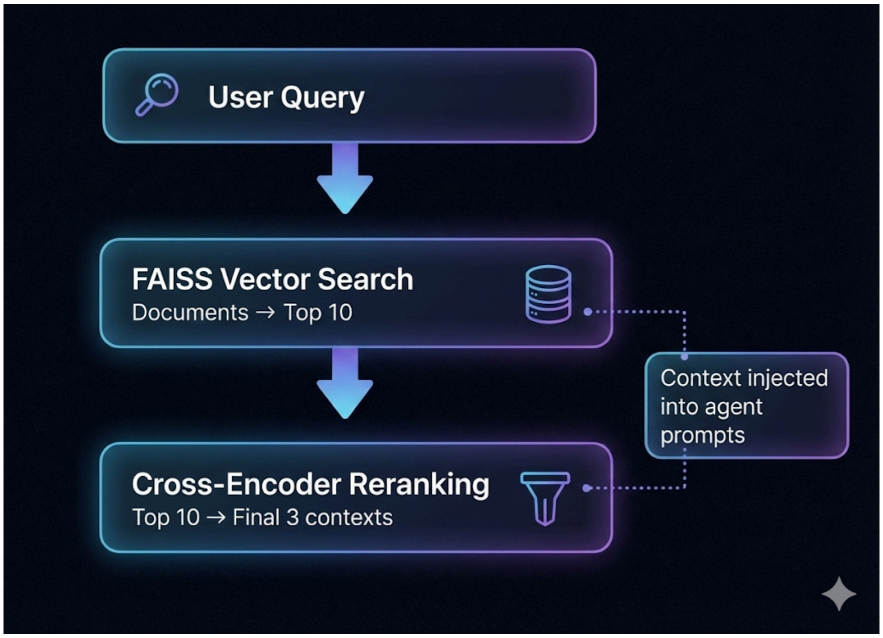

Figure 3: Two-Stage RAG Retrieval Process

5.4 Multi-Agent Design

Table 2: Multi-Agent Design - Roles, Metrics, and Outputs

Agent	Role	Key Metrics	Output
Fundamental	Financial health	P/E, ROE, margins	+1 / 0 / -1
Market Data	Price trends	52-week range	+1 / 0 / -1
Risk	Threat assessment	Volatility, competition	+1 / 0 / -1
Technical	Chart patterns	RSI, moving averages	+1 / 0 / -1
Chief	Synthesis	Sum of scores	BUY / SELL / HOLD
Statement	Document parsing	Ratios, metrics	6-section report

Scoring: Total ≥+2 = BUY, ≤-2 = SELL, else HOLD. LLM: Llama-3.3-70B (Groq).

5.5 User Interface & Deployment

The system features a web-based interface built with Gradio, providing two tabs: “Stock Analysis” for ticker queries and “Financial Statement Analysis” for document uploads. There is another tab “About” describing the project. The application is deployed on Hugging Face Spaces for free public access. The UI **(Figures 4-9)** is also compatible in both Dark and Light Modes.

Figure 4: Lumiq Intro Splash Screen

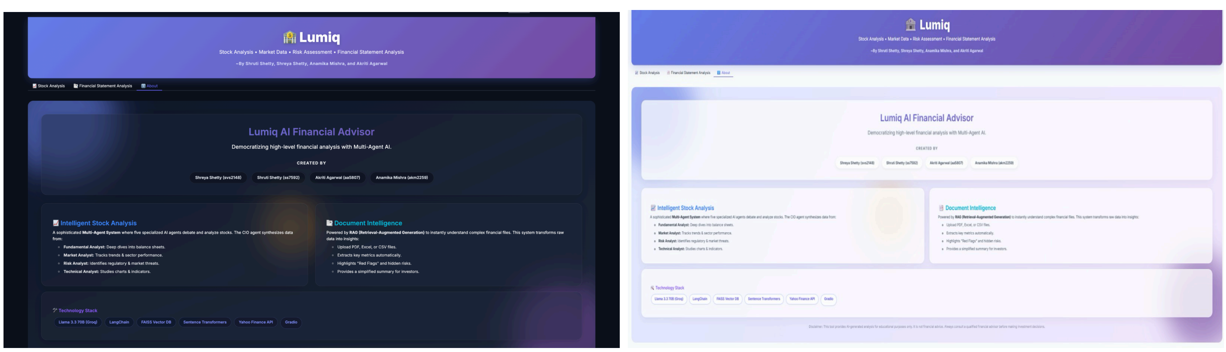

Figure 5: Lumiq About UI (Dark and Light Mode)

6. Evaluation / Results

6.1 Evaluation Approach

Lumiq's evaluation focuses on qualitative assessment of system outputs rather than quantitative metrics, as financial recommendation quality is inherently subjective and depends on market context, investor goals, and risk tolerance. We demonstrate the system's capabilities through two representative examples: stock analysis for Snap Inc. (SNAP) and financial statement analysis of Microsoft's 10-K filing.

6.2 Stock Analysis Example: Snap Inc. (SNAP)

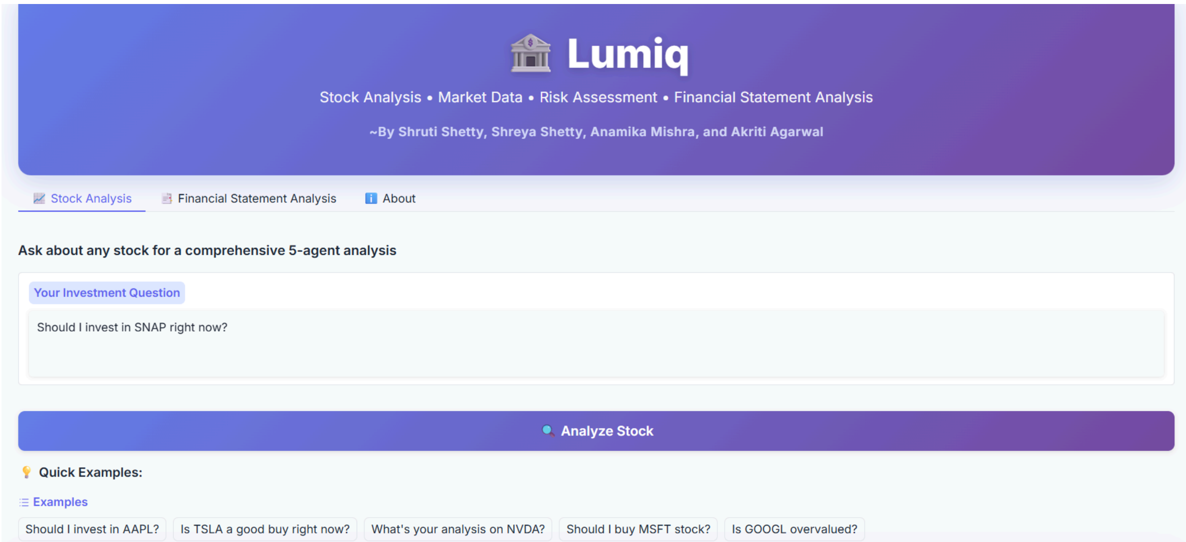

Figure 6: Lumiq Stock Analysis Interface

Figure 7: Lumiq Stock Analysis Report

This example (**Figure 6** and **Figure 7**) demonstrates balanced multi-perspective analysis where bearish signals from three agents outweighed neutral market positioning, resulting in a high-conviction sell recommendation with specific execution parameters.

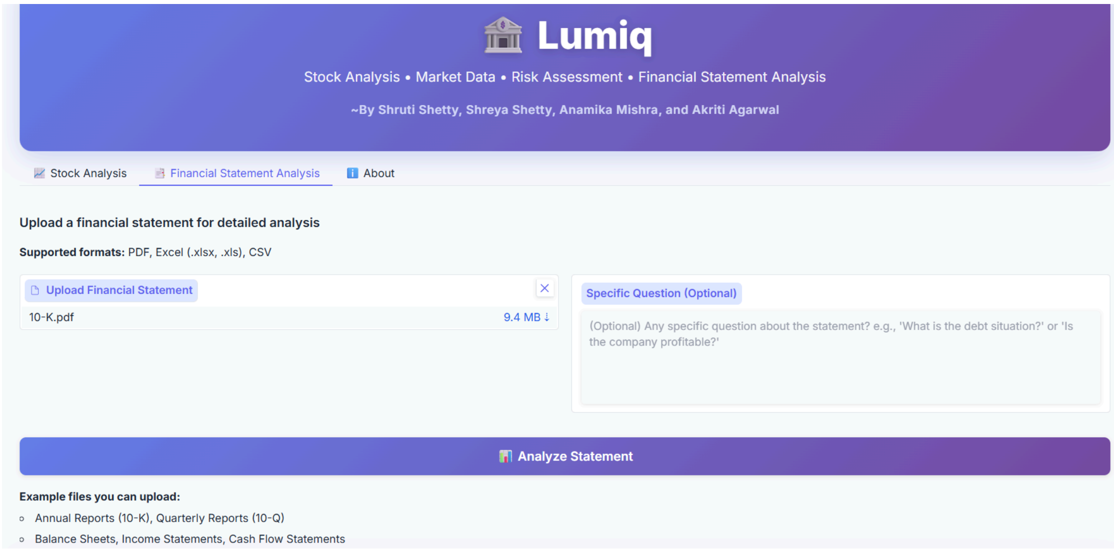

Figure 8: Lumiq Financial Statement Analysis Upload

Figure 9: Lumiq Financial Statement Analysis Report

In the above **Figures 8 and 9**, the financial statement parsing feature successfully extracts structured information and provides interpretive analysis across six dimensions.

7. DISCUSSION & LIMITATIONS

Lumiq successfully demonstrates a multi-agent architecture with diverse perspectives and RAG-based retrieval grounding responses in historical patterns and current market data, reducing hallucination risks. The transparent scoring mechanism (+1/0/-1) makes decision-making interpretable and educational. However, limitations include limited data coverage (128 documents, 10 live stocks), occasional financial statement parsing errors due to inconsistent formatting, lack of quantitative evaluation metrics or backtesting against actual market performance, and dependency on Groq API with potential quality degradation during rate-limiting fallback to smaller models. The system provides educational analysis without accounting for individual investor circumstances, tax implications, or macroeconomic factors.

8. CONCLUSION & FUTURE WORK

Lumiq demonstrates how multi-agent AI architecture democratizes financial analysis by providing diverse, interpretable investment perspectives through retrieval-augmented generation and specialized agent roles. By synthesizing recommendations from six agents into transparent conviction scores with detailed reasoning, the system makes institutional-quality analysis accessible for educational purposes.

Future enhancements include expanding data coverage through SEC EDGAR integration for automated 10-K/10-Q retrieval and earnings call transcripts, implementing backtesting to validate recommendation accuracy against historical performance, adding specialized agents for macroeconomic analysis and ESG scoring, and enabling personalization for user risk tolerance and portfolio context. Extending from point-in-time analysis to continuous monitoring with real-time alerts would provide ongoing investment support.